SPECIFICATION

Product Name: Methane Leakage Detector

Item No.: AM5303

Version: V2406061A-01

Date: June 6th, 2024

Revision

No.	Version	Content	Date
1	V2406061A-01	Preliminary version	2024.6.6
2			

Methane Leakage Detector

AM5303

Applications

- Ground pipe
- Abandoned wells and active wells in use

Description

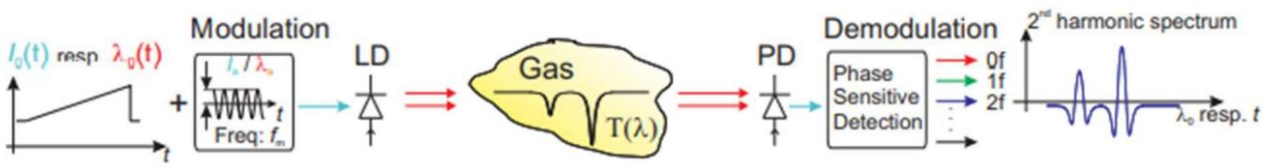
The methane leak detector is an industrial equipment used to monitor the concentration of methane gas in oil and gas wells, with characteristics such as field use, low power consumption, high-precision detection, and high resolution.

Features

- TDLAS methane sensor with independent intellectual property rights
- Fast response, high accuracy, long life span
- Adopting narrowband laser spectral absorption technology, it has unique selectivity for methane, no interference from other gases, water vapor, dust particles
- Measurement range: 0-1000ppm
- Long life time over 10 years
- Low power consumption

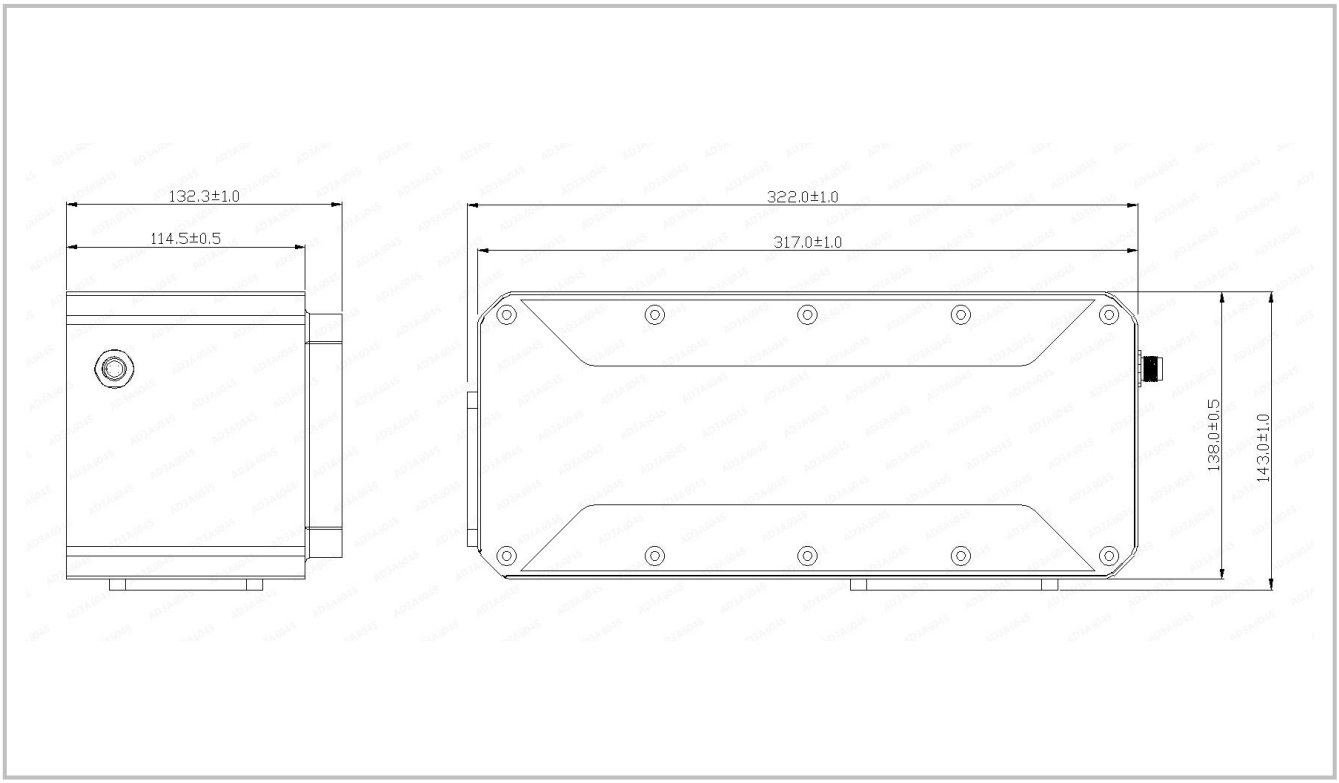
Working Principle

TDLAS methane sensor mainly consists of a laser light source, a gas chamber, a detector, etc. The laser source emits a specific wavelength of laser, and with the modulation of the injected periodic current, the wavelength undergoes periodic changes. Through wavelength scanning, the center wavelength of the laser output becomes the absorption spectral line of the gas to be measured, and the concentration of the gas to be measured is inverted using information such as spectral intensity signals obtained through gas absorption.

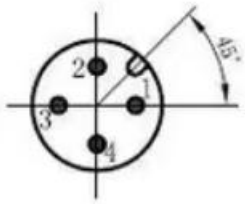


Product Dimensions and Pin Definition

1. Dimensions (Unit: mm)



2. Pin Definition

Pin No.	Function	Connector Pin (male, M12-B-P)
1 (Red)	VCC @ 12V	
2 (Black)	GND	
3 (Blue)	RS-485 A @ 12V	
4 (Green)	RS-485 B @ 12V	

Specifications

AM5303 Methane Leakage Detector Specification	
Target Gas	Methane (CH4)
Working Principle	Tunable Diode Laser Absorption Spectroscopy (TDLAS)
Measurement Range	0~1000 ppm
Accuracy [1)]	+/-(5 ppm + 2%reading) @0~40˚C; +/-(10 ppm + 2%reading) @-20~0˚C&40~50˚C;
Resolution	0.1 ppm
Detection Limit [2)]	1 ppm
Response Time	T90 ≤ 20s
Operation Mode [3)]	Continuous / Intermittent
Data Update Time [4)]	2s (instantaneous value) / 10s (average value)
Working Cycle [5)]	36s (in intermittent mode)
Dimension	143mm (W)* 322mm(L)* 132.3mm(H)
Output	RS485 @12V
Design Life Time	10 years
IP Grade	IP66
Certification	UL Class I Div 2
Working Temperature	-20 ~ 50℃
Working Humidity	0~100%RH (Non-condensing)
Working Pressure	86kPa- 106kPa
Working Place	Outdoor
Power Supply	DC 12V
Average Power Consumption	≤1W @10˚C, 24h

Note:

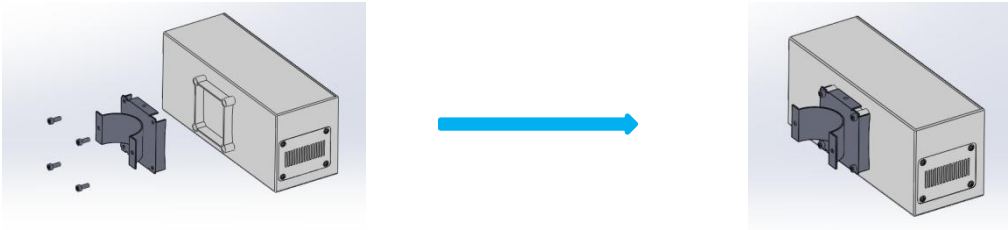
1) Accuracy is guaranteed in each piece-wise working temperature range and full humidity range under 1 atm.
2) The detection limit is defined as 3-Sigma of the CH4 readings in N2.
3) In continuous mode, the fan and sensor core are working continuously. In intermittent mode, the fan and the sensor core are working intermittently to save power consumption.
4) The detector has outputs for both instantaneous CH4 value (register 0x07) and averaged CH4 value (register 0x00). In continuous mode, the data update time of the instantaneous value is 2 seconds and that of the average value is 10 seconds.
5) Working cycle in intermittent mode is 36 seconds, during which the fan will work for 16 seconds then stop for 20 seconds, the sensor core will work for 26 seconds then stop for 10 seconds. Main controller of the detector starts collecting data from the sensor core from the 16th second and keeps for 10 seconds. The collected data will be averaged and output in register 0x00 in the Modbus protocol. See timing sequence at note 6 in section 3.5 of *Communication Protocol* chapter.

Installation Instruction

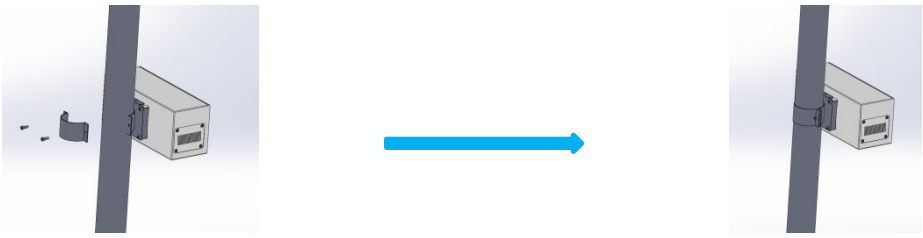
This product should be installed in a clean and dry environment, please do not install it when it is raining. In addition, installation parts are included in the package: solar panel installation, mounting components, column mounting plates and anti-theft screws.

1. Vertical installation

Step 1: Install the connecting ring onto the device



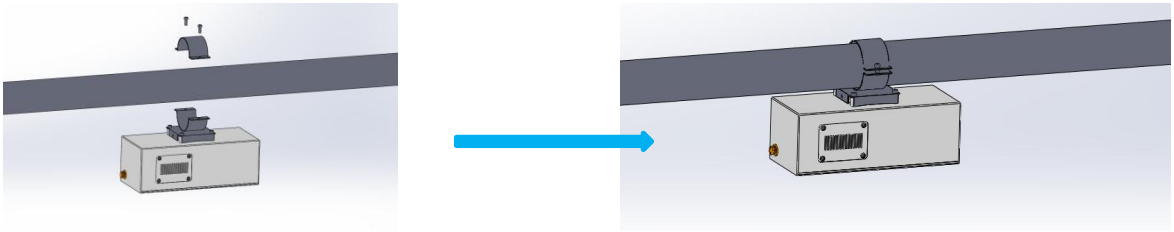
Step 2: Install the device on a vertical pole and secure it with a buckle;



2. Horizontal installation

Step 1: Install the connecting ring onto the device

Step 2: Install the device on a horizontal pole and secure it with a buckle;



Communication Protocol

1. Overview of Modbus Protocol

The communication protocol uses the standard Modbus protocol and Modbus-RTU transmission mode. During the communication process, the sensor always exists as a slave, responding to commands sent by the host.

2. Serial Port Configuration

Baud rate	Data bit	Stop bit	Check bit	Flow control
9600	8	1	0	None

The factory default baud rate is 9600.

3. Message Format

Device address	Function code	Data	Check
1 byte	1 byte	N byte	2 bytes

3.1. Function code description

No.	Function code	Operation	Explanation
1	0x03	Read Data Register	1 register contains 2 bytes
2	0x06	Write data register	1 register contains 2 bytes

3.2. Read Data Register (0x03)

Request		Response	
Field Description	(hexadecimal value)	Field Description	(hexadecimal value)
Device Address	0x01~0x84 or 0xC8	Device Address	0x01~0x84 or 0xC8
Function code	0x03	Function code	0x03
Starting address Hi	0x00~0xFF	Byte count	0x00~0xFF
Starting address Lo	0x00~0xFF	Number of Registers Hi	0x00~0xFF
Number of Registers Hi	0x00~0x80	Number of Registers Lo	0x00~0xFF
Number of Registers Lo	0x00~0x80	Number of Registers Hi	0x00~0xFF
CRC verification Lo	0x00~0xFF	Number of Registers Lo	0x00~0xFF
CRC verification Hi	0x00~0xFF	…	…
		…	…
		CRC verification Lo	0x00~0xFF
		CRC verification Hi	0x00~0xFF

3.3. Write Setting Register (0x06)

Request		Response	
Field Description	(hexadecimal value)	Field Description	(hexadecimal value)
Device Address	0x01~0x84 or 0xC8	Device Address	0x01~0x84 or 0xC8
Function code	0x06	Function code	0x06
Starting address Hi	0x00~0xFF	Starting address Hi	0x00~0xFF
Starting address Lo	0x00~0xFF	Starting address Lo	0x00~0xFF
Number of Registers Hi	0x00~0xFF	Number of Registers Hi	0x00~0x80
Number of Registers Lo	0x00~0xFF	Number of Registers Lo	0x00~0x80
CRC verification Lo	0x00~0xFF	CRC verification Lo	0x00~0xFF
CRC verification Hi	0x00~0xFF	CRC verification Hi	0x00~0xFF

3.4. Abnormal response

When the user is reading or writing a register, if the register address or value is illegal, an exception code will be returned.

Code	Name	Meaning
01	Illegal function code	Function code does not exist
02	Illegal register address	Register address out of range or invalid combination of register address and length
03	Illegal register value	The set register value is within the disallowed range

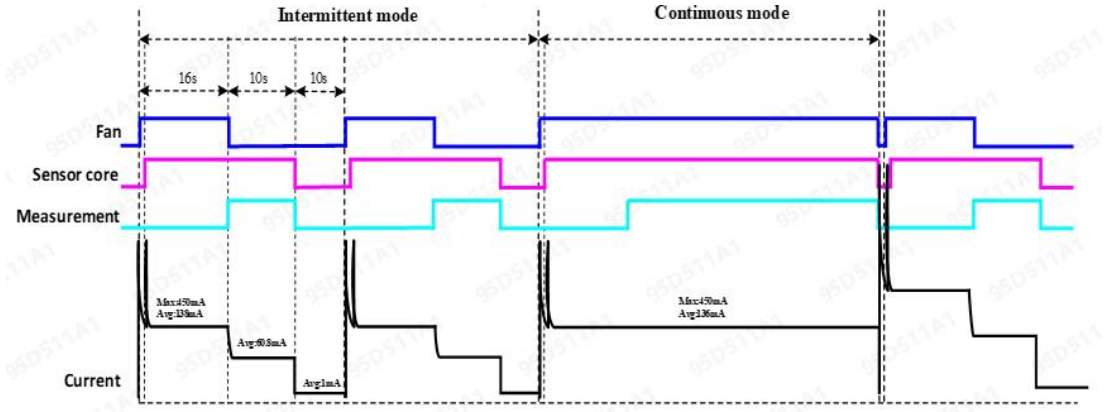
3.5. Register Detailed List

Register Address	Data Type	Byte Length	Read/Write	Description
0x00	short	2	Read	CH4 concentration (average)
0x01	short	2	Read	Sensor core temperature x100, unit: ℃
0x02	unsigned	2	Read	Sensor operation status code, see Appendix Table 4
0x03	unsigned	2	Read	Span
0x04	unsigned	2	Read	Number of decimal places, see *Appendix Table 1*
0x05	unsigned	2	Read	Concentration unit, see *Appendix Table 2*
0x06	unsigned	2	Read	Real time optical power
0x07	short	2	Read	CH4 concentration (instantaneous)
0x65	short	2	Read and write	Zero-point calibration
0x66	short	2	Read and write	Span point calibration
0x67	short	2	Write	Factory reset when writing 1
0x68	short	2	Read and write	Zeroing when writing 1
0x69	short	2	Read and write	Device Address
0x6A	short	2	Read and write	Baud rate, see *Appendix Table 3*
0x70	short	2	Read	Sensor model, fixed to 2502

0x71	short	2	Read	Material number, example: 1001
0x72	short	2	Read	Year of manufacture, e.g. 2023
0x73	short	2	Read	Factory month and day, for example: 0628
0x74	short	2	Read	Hardware version number, for example: 101 represents 1.0.1
0x75	short	2	Read	Software versioning, for example: 101 means 1.0.1
0x76	String	2	Read	SN1-SN2, sensor model, fixed to MS
0x77	String	2	Read	SN3-SN4
0x78	String	2	Read	SN5-SN6
0x79	String	2	Read	SN7-SN8
0x7A	String	2	Read	SN9-SN10
0x90	unsigned	2	Read and write	Work interval default: 36s the set value < 36s, continuous mode operation, and the average concentration value is updated every 10s; the set value ≥ 36, intermittent mode operation, and the average concentration value is updated every 36s.

Note:

1. 0xC8 is the default and universal device address for Modbus.
2. The device address can be set between 0x01 and 0x84.
3. The CH4 concentration, span, zero-point threshold, zero-point calibration and span point calibration are all magnified. The magnification is determined by the number of decimal places. See *Appendix Table 1* for details.
4. Factory reset will restore the calibration linearity to the factory state.
5. The sensor serial number needs to be converted to a decimal number.
6. Timing sequence diagram of the continuous mode and intermittent mode is as shown below:



Appendix Tables

1. Number of Decimal Places

Nr. of D.P.	Description
0	0000, which means the corresponding multiple is 1
1	000.0, which corresponds to a multiple of 10
2	00.00, which means the corresponding multiple is 100
3	0.000, which corresponds to a multiple of 1000

2. Concentration Unit

Unit	Unti Description
0	ppm
1	mg/m3
2	%

3. Baud Rate

Command	Description
0	Baud rate 9600 bps, factory default
1	Baud rate 19200 bps
2	Baud rate 38400 bps
3	Baud rate 115200 bps

4. Sensor Operation Status Code

Bit	0	1	2	3
meaning	Optical path failure	Abnormal temperature	Abnormal air pressure	Preheating
Bit	**4**	**5**	**6**	**7**
meaning	Temperature exceeds the measurement range	Abnormal calibration data	TEC temperature abnormality	signal saturation
Bit	**8**	**9**	**10**	**11**
meaning	Low optical power	Excessive humidity	Unstable Concentration	Abnormal fan speed
Bit	**12**	**13**	**14**	**15**
meaning	Heating up	Condensing	Reserved	Reserved

Note: The status code is 16-digit hexadecimal data, each 1 digit represents 1 alarm type, 0 means invalid, 1 means valid. There may be multiple states at the same time,

Example: 0x0A corresponds to binary 00000000 00001010, indicating that the sensor is warming up and temperature is abnormal; 0 corresponds to binary 00000000 00000000, indicating normal operation.

Fault description: The optical path failure may be caused by condensation, pollution, and occlusion on the mirror surface. Abnormal temperature and air pressure will lead to abnormal data correction which shall be checked before use. If the temperature exceeds the measuring range, it means that the working temperature exceeds the allowable temperature of the sensor, and the error may have deviation. If the calibration data is abnormal, restore the factory settings and recalibrate. The abnormal TEC temperature indicates that the internal temperature control is abnormal, and it is necessary to check whether the power supply meets the requirements.

Precautions

1. This product should not be violently impacted or knocked during transportation and installation, otherwise it may cause abnormal operation of the detector.
2. Please do not open the device and do not modify it in any way, including but not limited to replacing circuit boards and sensors.
3. This product must have the correct power supply. If the power supply is abnormal, the detector will not work.
4. Install this product in the correct way and do not install or operate the detector in any other direction.
5. This product is designed to be waterproof, rainproof or spray. Do not immerse the equipment in water if immersion waterproof is not designed.
6. Professional personnel should be regularly invited for maintenance, and any malfunctions must be repaired or replaced as soon as possible.
7. This product can reduce the occurrence of accidents, but it cannot guarantee safety.

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